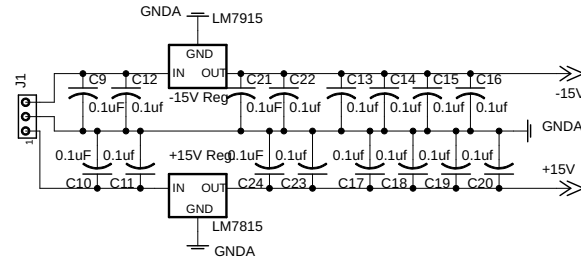


POWER SUPPLY:
Count with double Vdc = [18 - 24] Vdc.
All capacitors stabilize the voltage at input pin's from AO's and INA's.



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TITLE: NCC_GND_troubleshoot

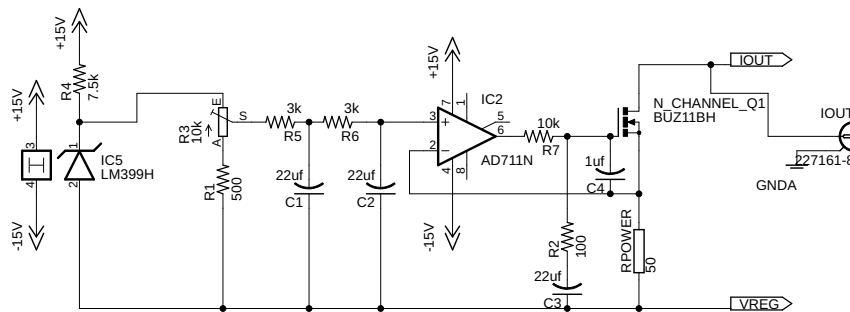
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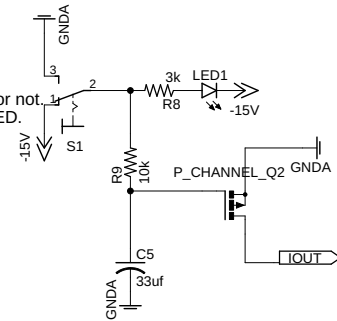
Date: Mayo 2024

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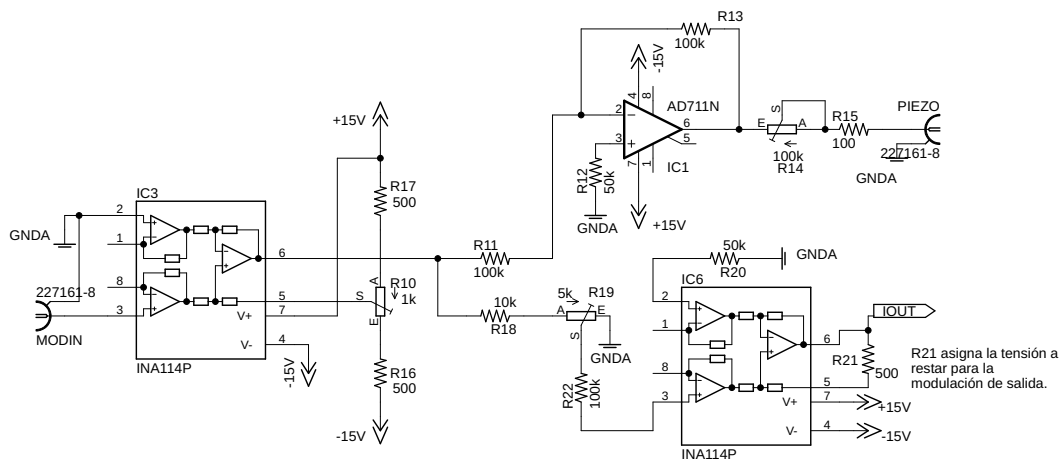
Reference voltage circuit based on LM399H (zener+thermal protection). Control the base current from Q1, allowing the sourcing current going trough the final element (laser diode).



General switch isolator circuit.
Allow the source current Q1 flow or not.
Indicates on state with inverted LED.
Q2 allows main supply to Q1.



Modulation circuit: It use 2 INA's to get a closed controlled loop based on trimmers R10 (offset), R19(voltage control gain), and R14 (piezo modulation voltage) signals.



Negative voltage reference power supply:
Main circuit for this board. It supplies to Q1 the needed current.
Includes a fast discharge circuit, being the max current 1.5A (LM337 limit).
The R25 trimmer controls the voltage referenced to the Rpower = 50[Ohms].
Depending the size of C7, the circuit rectifies the ripple effect on the current, because the voltage NEEDS to be the minimum possible (value recommender +700uF).

